SHEN JUNYI

Singapore
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EDUCATION

National University of Singapore Aug. 2024–Present
Ph.D. in Computer Science *Singapore*
Advisor: Prof. Yao Lu Research: LLM systems, agentic workflows, and inference optimization
President's Graduate Fellowship GPA: 4.75/5.00

Zhejiang University Sept. 2020–June 2024
Bachelor of Engineering in Automation *Hangzhou, China*
Minor in the Intensive Training Program of Innovation and Entrepreneurship, Chu Kochen Honors College
GPA: 3.98/4.00 Rank: 3/122

PUBLICATIONS

Peer-Reviewed Publications

Yuhao Shen, Tianyu Liu, Xinyi Hu, Jinyang Wu, **Junyi Shen**, Quan Kong, Zhang Baolin, Jun Zhang, Jun Dai, Shuang Ge, Chen Lei, Yue Li, Mingcheng Wan, and Cong Wang. *Draft Less, Retrieve More: Hybrid Tree Construction for Speculative Decoding* (GRAFT). NeurIPS 2026 MLForSys Workshop (Poster). Submitted to ACL ARR, August 2026.

Yuhao Shen, Tianyu Liu, **Junyi Shen**, Jinyang Wu, Quan Kong, Huan Li, and Cong Wang. *Double: Breaking the Acceleration Limit via Double Retrieval Speculative Parallelism.* ACL 2026 (Oral, Best Paper Candidate, SAC Highlight).

Noppanat Wadlom, **Junyi Shen**, and Yao Lu. *Efficient LLM Serving for Agentic Workflows: A Data Systems Perspective.* SIGMOD 2026 (accepted).

Yuhao Shen*, **Junyi Shen***, Quan Kong, Tianyu Liu, Yao Lu, and Cong Wang. *SpecBranch: Speculative Decoding via Hybrid Drafting and Rollback-Aware Branch Parallelism.* ICLR 2026 (Poster). *Equal contribution.

Preprints

Junyi Shen, Noppanat Wadlom, Zhengyuan Su, and Yao Lu. *DynBranch: Speculative Subgraph Reuse for Dynamic Agentic LLM Serving.* arXiv preprint, 2026. Submitted to EuroSys 2027.

Zhengyuan Su, Noppanat Wadlom, **Junyi Shen**, Peisong Zhang, Ran Luo, Dianbo Liu, Yi Hou, Yicong Huang, Wentao Wu, and Yao Lu. *Lumilake: An Agentic Analytics Engine for AI4Science.* CAIS '26 SAO Workshop (accepted). Submitted to VLDB 2027.

Junyi Shen, Noppanat Wadlom, Lingfeng Zhou, Dequan Wang, Xu Miao, Lei Fang, and Yao Lu. *FlowMesh: A Service Fabric for Composable LLM Workflows.* arXiv preprint, 2025.

Junyi Shen, Noppanat Wadlom, and Yao Lu. *Halo: Batch Query Processing and Optimization for Agentic Workflows.* arXiv preprint, 2025. Submitted to EDBT 2027.

Cheng Li, Ziang Leng, Chenxi Yan, **Junyi Shen**, et al. *ChatHaruhi: Reviving Anime Character in Reality via Large Language Model.* arXiv preprint, 2023.

RESEARCH EXPERIENCE

Research Intern, Shanghai Qi Zhi Institute Nov. 2023–May 2024
Advisors: Huanchen Zhang and Yao Lu
Developed continuous-batching techniques for Stable Diffusion serving.

Student Research Assistant, The University of Hong Kong July–Aug. 2023
Advisor: Hengshuang Zhao
Worked on cross-modal alignment and diffusion-model training.

HONORS

President's Graduate Fellowship, National University of Singapore 2024–Present
Outstanding Graduate of Zhejiang Province 2024
First-Class Scholarship, Zhejiang University 2021, 2022, 2023
National Scholarship, Ministry of Education of China 2021

TEACHING

Teaching Assistant, National University of Singapore
CS6216: Advanced Topics in Machine Learning AY 2025/26 Semester 1
CS4262/5462: Machine Learning Systems AY 2025/26 Semester 2

SELECTED TALK

MLSys Singapore Meetup #17 6 Nov. 2025
Batch Query Processing and Optimization for Agentic Workflows
National Supercomputing Centre Singapore

TECHNICAL SKILLS

Programming: Python, C/C++ **Tools:** PyTorch, Linux, Git, LaTeX, Codex, Claude Code

LEADERSHIP & SERVICE

League-Branch Secretary, Control 2001 / IT 2022 2020–2024
Executive Chairman, Student Union, College of Control Science and Engineering, Zhejiang University 2022
Committee Member, Student Committee, Zhejiang University 2022
Volunteer, 19th Asian Games, Hangzhou 2023